

Fleet Intelligence & Analytics

Turning fleet operations data into actionable insights.

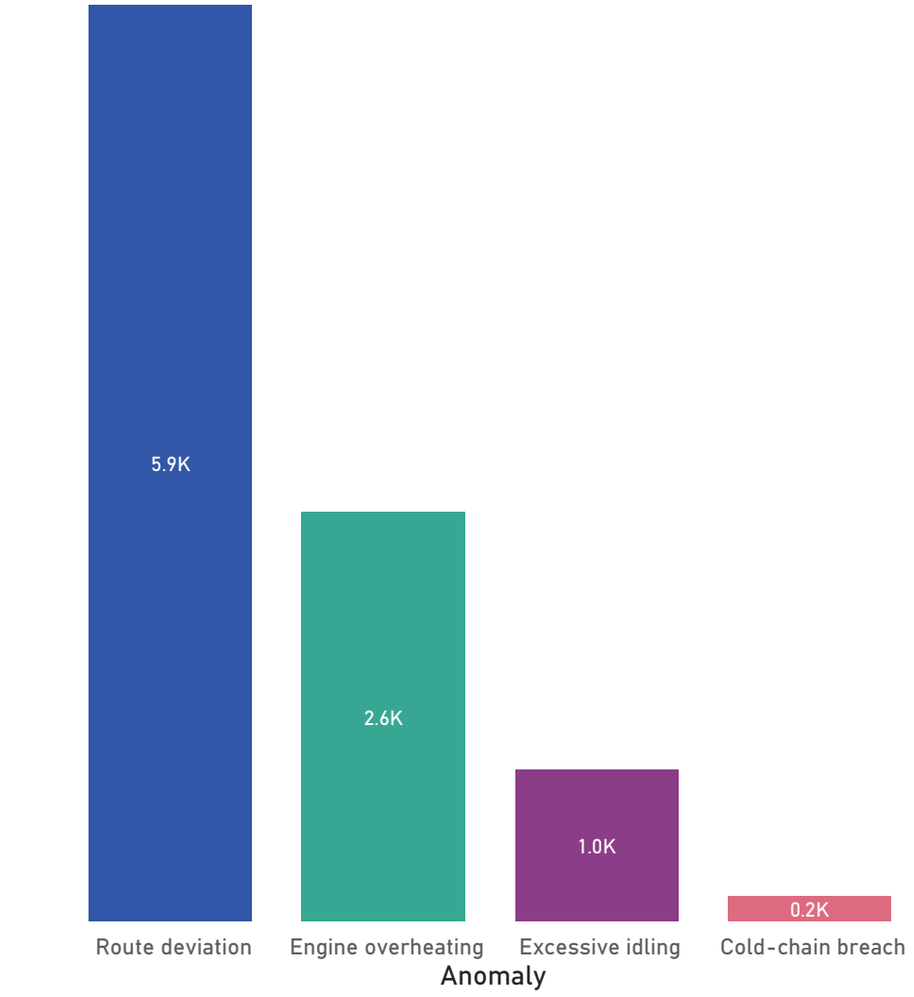
Monitor vehicle efficiency, route performance, driver behavior, cold-chain compliance, and data quality through one integrated analytics solution.

Fleet Overview

Target: 5,000

2.77	99.86%	163	10K	5K
Fuel Efficiency km/L	Cold Chain Compliance %	Cold Chain Breach Episodes	Anomaly Episodes	Vehicles Reporting

Anomaly Episodes by Type



Route Performance Summary

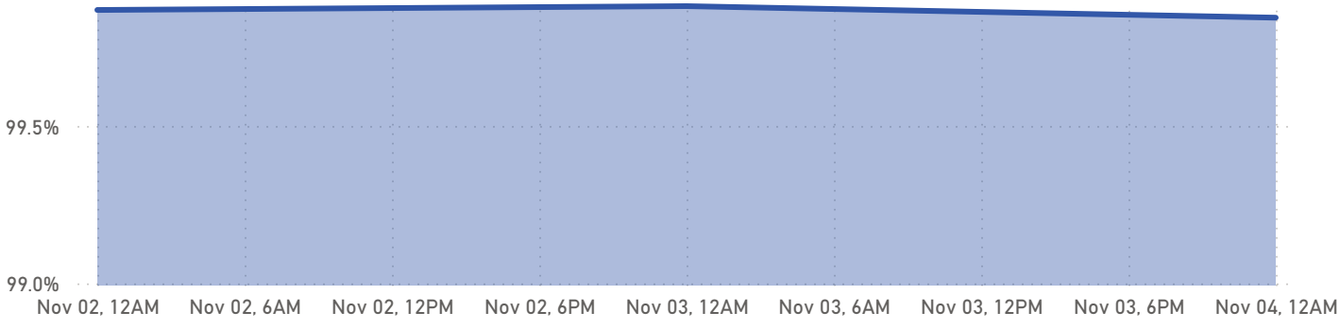
Route	Fuel Efficiency vs Fleet km/L	Cold Chain Compliance	Episodes per 1,000 km
Alexandria - Hurghada	0.00	99.88%	
Hurghada - Mansoura	0.00	99.82%	
Asyut - Cairo	0.00	99.79%	
Hurghada - Minya	0.00	99.78%	
Cairo - Suez	0.00	99.86%	
Ismailia - Port Said	0.00	99.73%	
Asyut - Hurghada	0.00	99.90%	
Mansoura - Minya	0.00	99.91%	
Asyut - Port Said	0.00	99.84%	
Hurghada - Port Said	0.00	99.86%	
Alexandria - Minya	0.00	99.87%	
Cairo - Ismailia	0.00	99.82%	
Ismailia - Mansoura	0.00	99.88%	
Alexandria - Ismailia	0.00	99.87%	
Minya - Tanta	0.00	99.81%	
Cairo - Port Said	0.00	99.85%	
Asyut - Minya	0.00	99.87%	
Alexandria - Mansoura	0.00	99.87%	
Alexandria - Tanta	0.00	99.90%	
Port Said - Suez	0.00	99.89%	
Alexandria - Asyut	0.00	99.97%	
Cairo - Minya	0.00	99.85%	
Minya - Port Said	0.00	99.84%	
Total	**0.00**	**99.86%**	**1.82**

Routes and Cold Chain

Cold-Chain Breach Episodes by Route



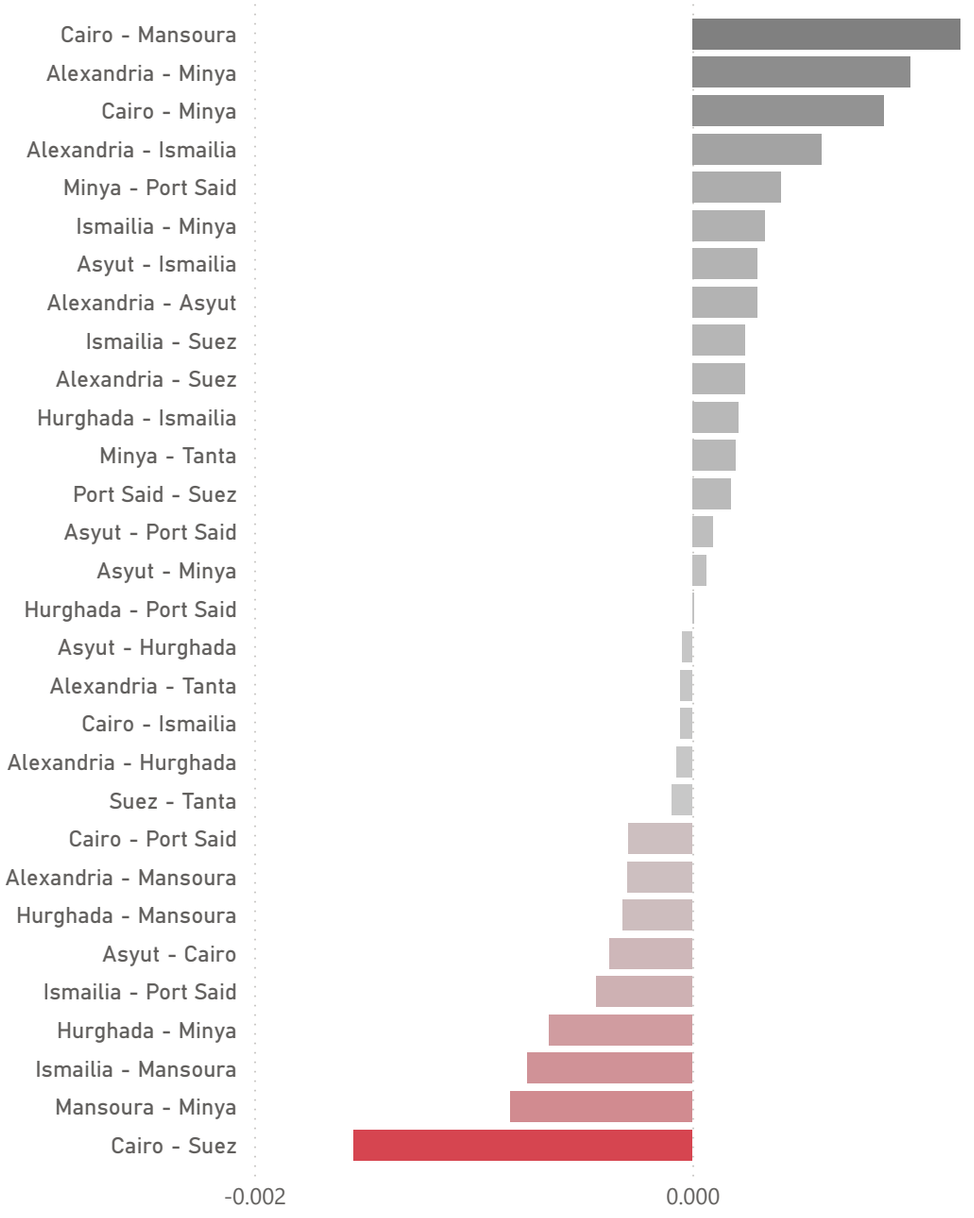
Cold-Chain Compliance Trend



Route Performance Details

Route	Total km	Total Litres	Cold Readings Total	Cold Chain Compliance %	Cold Chain Breach Episodes
Minya - Port Said	151,237.09	54,651.00	92819	99.84%	11
Asyut - Port Said	212,994.07	76,976.21	108629	99.84%	10
Hurghada - Mansoura	215,735.34	77,978.41	99922	99.82%	10
Hurghada - Port Said	203,520.07	73,554.61	100476	99.86%	10
Minya - Tanta	229,415.01	82,907.64	120123	99.81%	10
Hurghada - Minya	233,566.11	84,433.66	109432	99.78%	9
Alexandria - Mansoura	170,431.51	61,602.63	58464	99.87%	7
Asyut - Cairo	149,920.45	54,190.51	61928	99.79%	7
Total	**5,345,177.79**	**1,931,814.52**	**2415864**	**99.86%**	**163**

Fuel Efficiency by Route (km/L)



Driver Scorecard

20.58K	983	8.63%	6K
Idling Minutes	Excessive Idling Episodes	Speeding Rate %	Route Deviations

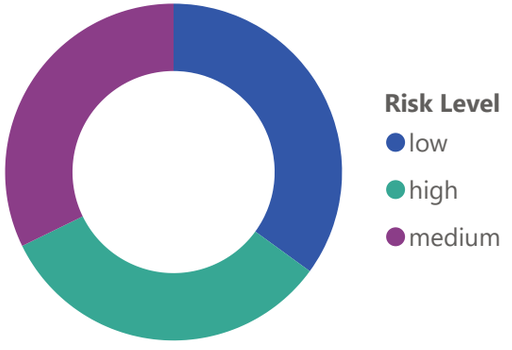
Driver Performance Details (Top 20 By Idling Minutes)

Driver Name	Home Depot	Risk Level	Idling Minutes	Idle Share of Trip Time	Excessive Idling Episodes	Speeding Rate	Route Deviations
Ali Kamel	DEP-ISM	low	56.50	9.75%	2	7.34%	0
Ahmed Hassan	DEP-TNT	medium	52.50	8.24%	2	8.77%	1
Tarek Farouk	DEP-MNS	low	58.00	7.30%	3	7.56%	1
Khaled Kamel	DEP-CAI	medium	53.00	7.27%	2	8.01%	0
Sara Farouk	DEP-MNY	high	54.00	6.79%	2	8.87%	0
Sara Fouad	DEP-HRG	low	57.00	6.07%	2	8.24%	0
Ali Nabil	DEP-ISM	low	58.00	5.53%	2	7.51%	2
Dina Zaki	DEP-SUZ	low	53.00	5.18%	2	8.30%	0
Tarek Ramadan	DEP-MNY	high	53.50	4.76%	2	8.26%	1
Ahmed Zaki	DEP-MNY	medium	55.00	4.62%	2	8.31%	0
Sherif Abdelrahman	DEP-MNY	low	52.00	4.60%	1	8.54%	0
Sherif Nabil	DEP-HRG	medium	51.00	4.57%	2	8.41%	0
Khaled Shawky	DEP-AST	medium	56.50	4.54%	2	7.70%	2
Ali Fouad	DEP-HRG	medium	53.00	4.45%	2	8.59%	3
Khaled Ramadan	DEP-ISM	low	53.50	4.33%	2	7.88%	1
Mostafa Salem	DEP-HRG	low	51.50	4.20%	2	9.16%	0
Walid Ramadan	DEP-ISM	low	52.00	4.12%	2	8.89%	2
Karim Shawky	DEP-HRG	low	54.50	4.11%	2	8.84%	1
Amr Shawky	DEP-HRG	low	56.50	3.93%	2	9.86%	0
Heba Fathy	DEP-HRG	medium	54.00	3.83%	2	8.15%	2
Sara Fouad	DEP-HRG	high	0.00	0.00%	0	9.66%	0
Total			**1,085.00**	**4.91%**	**40**	**8.45%**	**16**

Home Depot

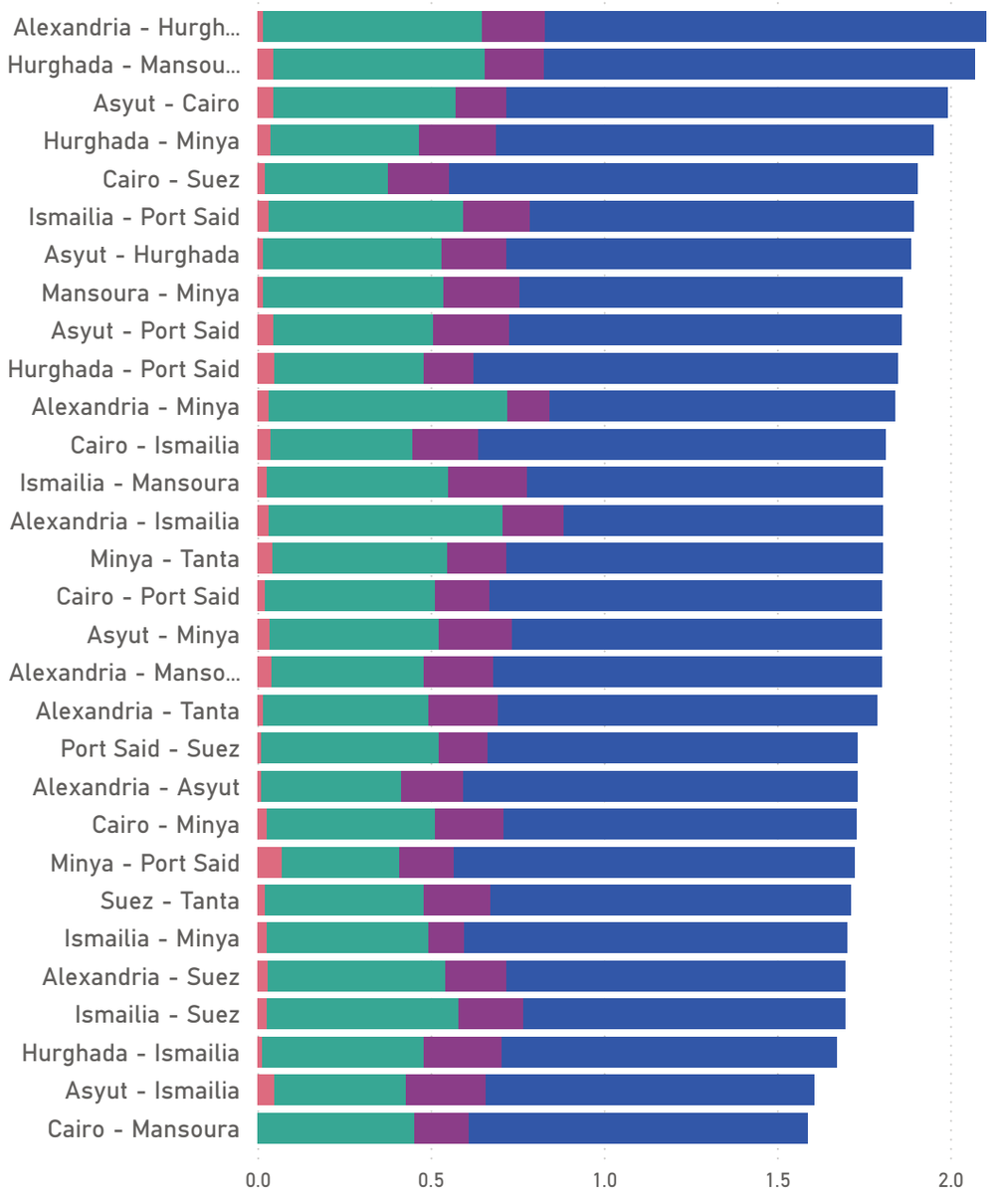
DEP-ALX | DEP-MNS
DEP-AST | DEP-MNY
DEP-CAI | DEP-PSD
DEP-HRG | DEP-SUZ
DEP-ISM | DEP-TNT

Idle Share by Risk Level

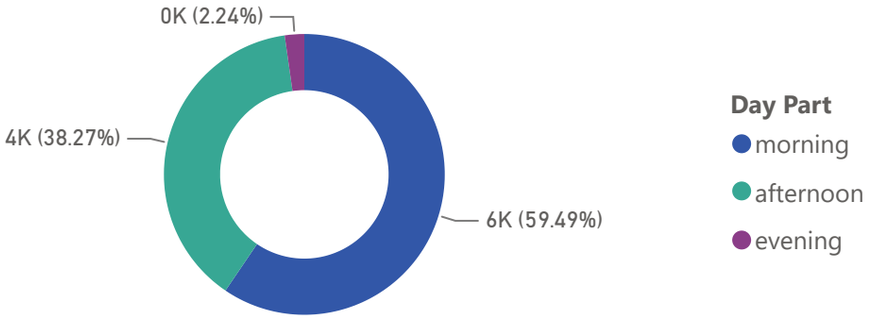


Anomaly Investigation

Episodes per 1,000 km by Route and Anomaly Type



Episodes by Time of Day



Average Episode Duration Minutes by Type



Anomaly

- Cold-chain breach
- Engine overheati...
- Excessive idling
- No anomaly
- Route deviation

Vehicle Type

- dry_van
- refrigerated_truck

Detected Episode Details (Top 20 By Sum of Duration Minutes)

Episode	Vehicle	Route	Anomaly	Month	Day	Duration Minutes	Reading Count
VEH-03073|excessive_idling|1	VEH-03073	Mansoura - Minya	Excessive idling	November	3	66.00	104
VEH-01650|excessive_idling|1	VEH-01650	Asyut - Minya	Excessive idling	November	2	35.00	62
VEH-03768|excessive_idling|1	VEH-03768	Alexandria - Hurghada	Excessive idling	November	3	32.00	54
VEH-00004|excessive_idling|1	VEH-00004	Cairo - Suez	Excessive idling	November	2	29.50	60
VEH-	VEH-00283	Cairo - Ismailia	Excessive idling	November	2	29.50	60
Total						**1,077.00**	**2140**

Pipeline Health

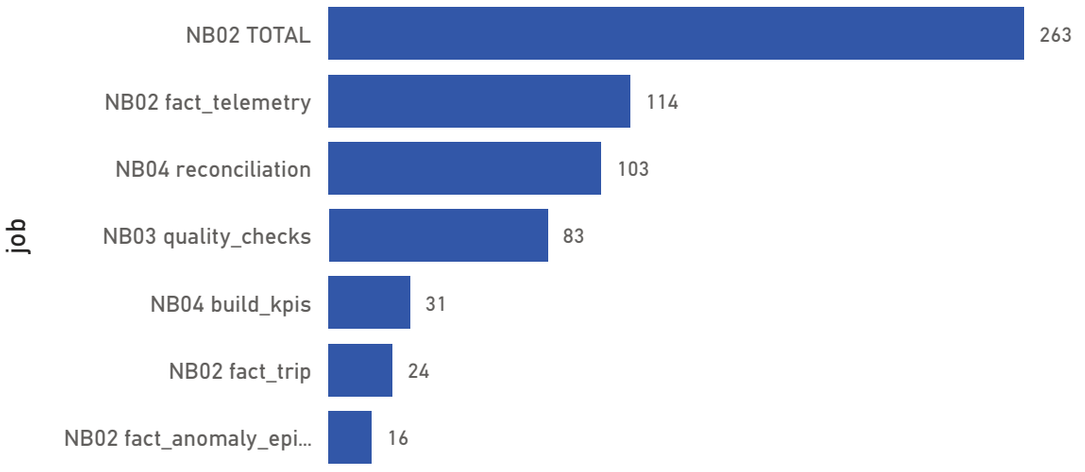
Expected 416

28	1	--	100.00%	416
Checks PASS (latest run)	Checks WARN (latest run)	Checks FAIL (latest run)	Recon Match %	Drivers With SCD2 History

Data Quality Results

check_name	gold_dim_driver	gold_fact_anomaly_episode	gold_fact_telemetry	gold_fact_trip
fk_anomaly_type_key		0	0	
fk_date_key			0	
fk_driver_key		0	0	0
fk_route_key		0	0	0
fk_start_date_key		0		0
fk_time_key			0	
fk_vehicle_key		0	0	0
no_duplicate_rows		0	0	0
no_null_keys		0	0	0
one_current_row_per_driver	0			
range_gold_measures		0	2797	0
range_sensor_values			32638	
row_count_matches_silver		0	93408	0

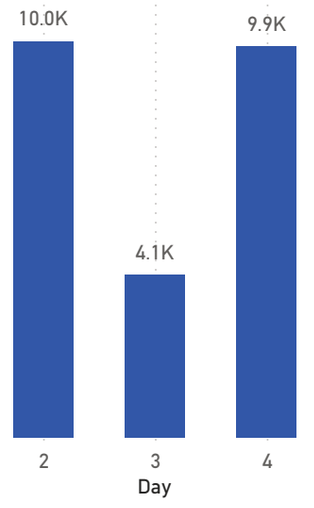
Execution Duration by Job



Silver-to-Gold Reconciliation Details

kpi	metric	Sum of groups_compared	Sum of groups_mismatched	Sum of gold_total	Sum of silver_total	Sum of max_abs_diff	status
cold_chain_compliance	compliance_pct	30	0	2,995.83	2,995.83	0.00	MATCH
cold_chain_compliance	overall_compliance_pct	1	0	99.86	99.86	0.00	MATCH
cold_chain_compliance	readings_in_range	30	0	2,412,490.00	2,412,490.00	0.00	MATCH
cold_chain_compliance	readings_total	30	0	2,415,864.00	2,415,864.00	0.00	MATCH
driver_scorecard	excessive_idling_episodes	13184	0	983.00	983.00	0.00	MATCH
driver_scorecard	idling_minutes	13184	0	20,582.00	20,582.00	0.00	MATCH
driver_scorecard	route_deviations	13184	0	5,932.00	5,932.00	0.00	MATCH
driver_scorecard	speeding_readings	13184	0	695,170.00	695,170.00	0.00	MATCH
fuel_efficiency_per_route	km_per_litre	30	0	83.01	83.01	0.00	MATCH
fuel_efficiency_per_route	total_km	30	0	5,345,177.79	5,345,177.79	0.00	MATCH
fuel_efficiency_per_route	total_litres	30	0	1,931,814.52	1,931,814.52	0.00	MATCH
Total		**52917**	**0**	**12,831,192.01**	**12,831,192.01**	**0.00**	

Daily Data Volume



Definitions & Methodology

1. Fuel Efficiency

Definition: Total distance travelled divided by total fuel consumed.
Calculation: SUM(total_km) / SUM(total_litres)
Data handling: Intervals longer than 300 seconds and GPS jumps are excluded from the relevant calculations.
Unit: Kilometres per litre (km/L).

2. Cold-Chain Compliance

Definition: Refrigerated sensor readings within the simulation's acceptable temperature range divided by all refrigerated readings.
Calculation: Readings within 2–8 °C / All refrigerated readings * 100
Threshold: 2–8 °C, inclusive.
Limitation: This metric measures adherence to the simulation threshold. It is not evidence of regulatory compliance or certification.

3. Speeding and Idling

Speeding: A sensor reading with a speed greater than 90 km/h.
Threshold: Speed > 90 km/h
Idling: A reading where vehicle speed is below 2 km/h and engine RPM is greater than zero.
Threshold: Speed < 2 km/h AND RPM > 0 Planned stops with RPM equal to zero are excluded from the idling definition.

4. Anomalies and Episode Counts

Anomaly, speeding, idling, route-deviation, and cold-chain breach counts represent **detected events or episodes**, according to the implemented detection rules. These counts are not independently verified ground truth. A detected event does not necessarily establish that an actual operational incident occurred.

5. Data Window and Limitations

Reporting window: Use the actual start and end timestamps represented in the Gold-layer data.
Data freshness: The latest available trip-end timestamp is displayed as the report's *Data Through* value.
Partial-day note: The latest reporting day may contain incomplete data. Daily totals and rates for that day may therefore be lower or less representative than those for complete days.
Interpretation: Results depend on available sensor data, implemented filtering rules, and simulation assumptions. Interpret comparisons with these limitations in mind.